

Physics of the Earth as a Planet
Course Syllabus – Michaelmas term 2026
Monday, Wednesday, Friday – 9 am – Ray Dolby Centre, Lecture Theatre

October 9	Introduction – Gualandi
October 9	Reception for Part III students – 4:00 pm Tea room – Madingley Rise House
October 12	Plate tectonics – Gualandi
October 14	Magnetic anomalies and past plate motions – Gualandi
October 16	Gravity, isostasy and creep – Gualandi
October 19	Thermal structure of plates – Gualandi
October 21	Plate flexure – Gualandi
October 23	Continuum mechanics – Rudge
October 26	Postglacial uplift – Rudge
October 28	Fluid dynamics of mantle convection – Rudge
October 30	Approximate boundary layer analysis – Rudge
November 2	Forces on plates maintaining their motion – Rudge
November 4	Planetary geophysics – Rudge
November 6	Finite elasticity – Al-Attar
November 9	Constitutive theory – Al-Attar
November 11	Seismic sources – Al-Attar
November 13	Plane wave propagation – Al-Attar
November 16	Ray theory – Al-Attar
November 18	Spherical Earth structure – Al-Attar
November 20	Delay time tomography – Al-Attar
November 23	Waveform tomography – Al-Attar
November 25	Self-gravitation and rotation – Al-Attar
November 27	Equilibrium figures – Al-Attar
November 30	Free oscillations – Al-Attar
December 2	Mode splitting and coupling – Al-Attar

Supplemental reading:

C.M.R. Fowler, *The Solid Earth*, second edition, CUP, 2004.

F.A. Dahlen and J. Tromp, *Theoretical Global Seismology*, Princeton, 1998.

J.A. Hudson, *The excitation and propagation of elastic waves*, CUP, 1980.

D.L. Turcotte and G. Schubert, *Geodynamics*, third edition, CUP, 2014.